Confidently Wrong: Measuring and Mitigating Calibration Risks in LLMs for African Languages

Supplementary Appendix

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This appendix accompanies the anonymous submission "Confidently Wrong: Measuring and Mitigating Calibration Risks in LLMs for African Languages" and is provided as a separate file, in compliance with the Trustworthy AI Workshop (Deep Learning Indaba 2026) submission guidelines. Section, table, and figure numbers refer to this appendix; "the main paper" refers to the accompanying extended abstract.

Appendix A. Models

Table 1: Open-weight models evaluated.

Model	Params	Type	Hugging Face ID
Qwen3-4B-Instruct	4B	Instruction-tuned	`Qwen/Qwen3-4B-Instruct`
Gemma-4-12B	12B	Instruction-tuned	`google/gemma-4-12b-it`
Aya-Expanse-8B	8B	Multilingual	`CohereLabs/aya-expanse-8b`
AfroLlama-V1	7B	Africa-specific (LLaMA-2 base)	`Jacaranda/AfroLlama_V1`

AfroLlama-V1 is based on LLaMA-2-7B and was fine-tuned on Swahili, Zulu, Yoruba, Hausa, Xhosa, and English; Amharic and N. Sotho, which appear in our benchmarks, are out-of-distribution for this model. All models are loaded in `bfloat16` with scaled dot-product (`sdpa`) attention, a 2048-token context limit, and greedy decoding (temperature 0), on a single NVIDIA A100 (40 GB).

Appendix B. Metric definitions

For an item with options o and gold answer y, the cloze score of option o with tokens $o_1, \ldots, o_L$ given context x is the length-normalised log-probability

$$s(o) = \frac{1}{L} \sum_{t=1}^{L} \log p(o_t \mid x, o_{<t}). \tag{1}$$

$$p(o) = \mathrm{softmax}_o \, s(o), \qquad \hat{y} = \arg\max_o p(o), \qquad c = \max_o p(o). \tag{2}$$

Over N items with correctness $y_i \in \{0, 1\}$, confidence c_i, and $B = 15$ equal-width confidence bins $\{B_b\}$:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left| \mathrm{Acc}(B_b) - \mathrm{conf}(B_b) \right|. \tag{3}$$

$$\text{Overconfidence} = \frac{1}{N}\sum_i c_i - \frac{1}{N}\sum_i y_i. \tag{4}$$

$$\text{Brier} = \frac{1}{N}\sum_i (c_i - y_i)^2. \tag{5}$$

AUROC is the area under the ROC curve using c to rank correct against incorrect items. AUARC is the area under the accuracy-rejection curve: items are answered most-confident-first and accuracy on the answered subset is integrated over coverage. Temperature scaling rescales the logits by a single $T > 0$, $p_T(o) = \text{softmax}_o\big(s(o)/T\big)$; we fit $T^\star = \arg\min_T \text{ECE}$ on a seeded 50% calibration split and report ECE on the held-out half.

Appendix C. Prompts

Each option is scored as a continuation of the prompt; the prompt lists no answer letters.

Uhura-TruthfulQA (truthfulness)

```
Q: {question}
A:
```

AfriMMLU (knowledge)

```
The following is a multiple-choice question about {subject}. Give the
correct answer.
Question: {question}
Answer:
```

For 5-shot prompting, the prompt is prefixed with five solved examples in the same language, each formatted as the template above followed by its gold answer text and separated by blank lines; the examples are held out from evaluation.

Appendix D. Interventions

By *interventions* we mean lightweight, post-hoc procedures applied at or after inference to improve calibration or to trade answer coverage for reliability, without any retraining of the model. We evaluate three:

1. **Temperature scaling.** We fit one temperature T per language by minimising ECE on a seeded 50% calibration split and evaluate on the held-out half, searching the grid $T \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0\}$.
2. **Selective abstention.** The model answers only when its calibrated confidence exceeds a per-language threshold chosen from the risk-coverage curve; we report AUARC integrated over all coverage levels.
3. **Few-shot prompting.** Five in-context examples per language, drawn from held-out items, formatted identically to the evaluation prompt and prepended to each query.

Appendix E. Deployment framework

Figure 1 translates our findings into a deployment policy for African-language LLMs. The model's raw confidence is first recalibrated by per-language temperature scaling; the calibrated confidence $\hat{c}$ is then compared against two per-language thresholds. High-confidence queries ($\hat{c} \geq \tau_{\text{hi}}$) are answered, medium-confidence queries trigger a request for more context and a re-query, and low-confidence queries ($\hat{c} < \tau_{\text{lo}}$) are abstained on or deferred to a human; all low-confidence queries are logged to guide future African-language data collection.

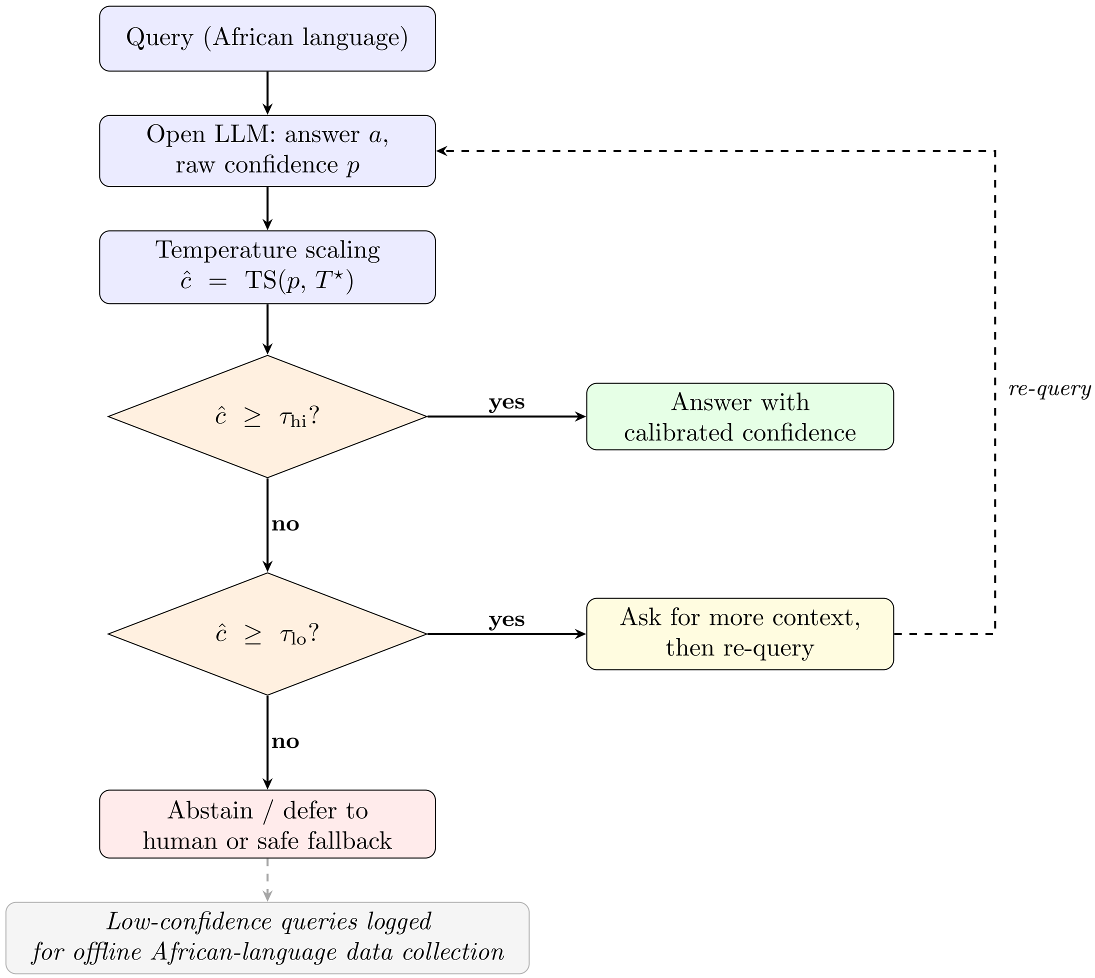


Figure 1: Deployment framework for LLMs in African languages. Recalibrate first with per-language temperature scaling, then route on calibrated confidence $\hat{c}$: high values ($\hat{c} \geq \tau_{\text{hi}}$) are answered with the stated confidence; medium values trigger a request for more context followed by re-query; low values ($\hat{c} < \tau_{\text{lo}}$) are abstained or deferred to a human or safe fallback. Thresholds τ are set per language from the risk-coverage curve, and low-confidence queries are logged to drive offline African-language data collection.

Appendix F. Additional results

Table 2: Per-language accuracy and ECE on Uhura-TruthfulQA (zero-shot). Each cell shows Acc / ECE; **bold** ECE exceeds the English baseline for that model. † out-of-distribution for AfroLlama-V1.

Language	AfroLlama-V1	Aya-Expanse-8B	Gemma-4-12B	Qwen3-4B
English (eng)	0.292 / 0.177	0.358 / 0.075	0.329 / 0.212	0.381 / 0.072
Amharic (amh)†	0.294 / 0.101	0.302 / 0.077	0.274 / **0.293**	0.373 / 0.042
Hausa (hau)	0.363 / 0.085	0.302 / **0.235**	0.397 / **0.223**	0.308 / **0.199**
N. Sotho (nso)†	0.246 / **0.307**	0.260 / **0.293**	0.379 / **0.232**	0.270 / **0.262**
Swahili (swa)	0.341 / 0.122	0.314 / **0.215**	0.328 / **0.273**	0.335 / **0.157**
Yoruba (yor)	0.371 / 0.072	0.351 / **0.141**	0.329 / **0.256**	0.310 / **0.150**
Zulu (zul)	0.337 / 0.122	0.319 / **0.237**	0.343 / **0.266**	0.324 / **0.204**

Table 3: Per-language accuracy and ECE on AfriMMLU (zero-shot), representative subset. Each cell shows Acc / ECE; **bold** ECE exceeds the English baseline for that model.

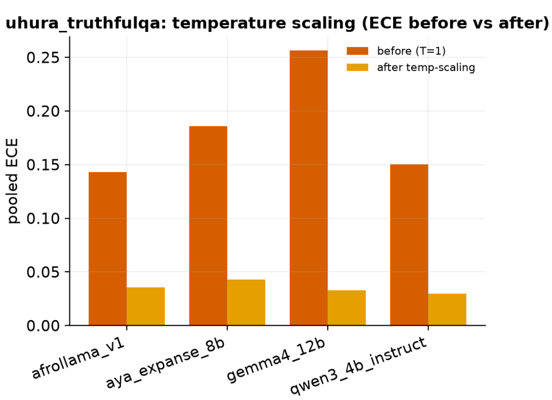
Language	AfroLlama-V1	Aya-Expanse-8B	Gemma-4-12B	Qwen3-4B
English (eng)	0.356 / 0.225	0.520 / 0.057	0.296 / 0.311	0.526 / 0.062
Amharic (amh)	0.258 / 0.147	0.240 / **0.114**	0.268 / **0.314**	0.286 / **0.084**
Hausa (hau)	0.294 / 0.190	0.308 / **0.172**	0.308 / **0.309**	0.304 / **0.141**
Swahili (swa)	0.294 / 0.188	0.312 / **0.151**	0.280 / **0.325**	0.312 / **0.139**
Yoruba (yor)	0.290 / 0.192	0.300 / **0.154**	0.250 / **0.349**	0.308 / **0.126**
Zulu (zul)	0.310 / 0.171	0.292 / **0.173**	0.252 / **0.362**	0.286 / **0.167**
Ewe (ewe)	0.244 / **0.239**	0.274 / **0.211**	0.252 / **0.374**	0.288 / **0.175**
Igbo (ibo)	0.252 / **0.245**	0.262 / **0.214**	0.288 / **0.329**	0.258 / **0.202**
French (fra)	0.280 / **0.268**	0.442 / **0.071**	0.244 / **0.383**	0.372 / **0.114**

Table 4: Per-language temperature scaling on Uhura-TruthfulQA for AfroLlama-V1 and Qwen3-4B. $T^\star$ is selected on the calibration half; ECE values are on the held-out half.

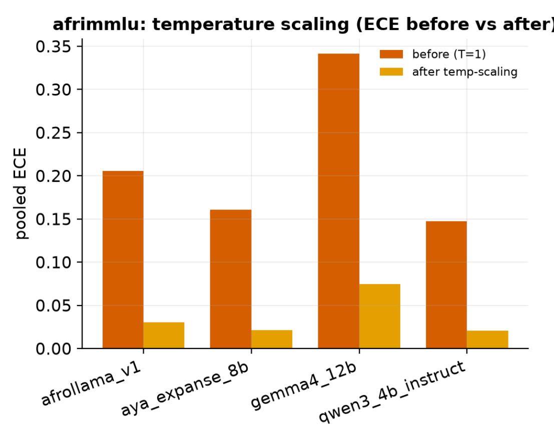
	AfroLlama-V1			**Qwen3-4B**		
Language	$T^\star$	ECE(bef)	ECE(aft)	$T^\star$	ECE(bef)	ECE(aft)
English	5.0	0.175	0.069	1.5	0.096	0.074
Amharic	5.0	0.115	0.038	1.2	0.056	0.043
Hausa	2.0	0.085	0.077	5.0	0.203	0.027
N. Sotho	5.0	0.307	0.093	5.0	0.260	0.067
Swahili	2.0	0.160	0.074	3.0	0.182	0.043
Yoruba	1.5	0.078	0.064	5.0	0.133	0.062
Zulu	2.0	0.126	0.046	5.0	0.191	0.085
Pooled	n/a	0.143	0.036	n/a	0.151	0.030

Table 5: Zero-shot versus 5-shot calibration on Uhura-TruthfulQA (values shown as zero-shot → 5-shot, macro-averaged over languages). Few-shot gives small gains for three models and backfires on Gemma-4-12B.

Model	**Accuracy**	**ECE**	**Overconfidence**
Qwen3-4B-Instruct	0.329 → 0.355	0.155 → 0.141	0.151 → 0.130
AfroLlama-V1	0.320 → 0.339	0.141 → 0.119	0.137 → 0.113
Aya-Expanse-8B	0.315 → 0.331	0.182 → 0.171	0.179 → 0.164
Gemma-4-12B	0.340 → 0.305	0.251 → 0.332	0.250 → 0.331

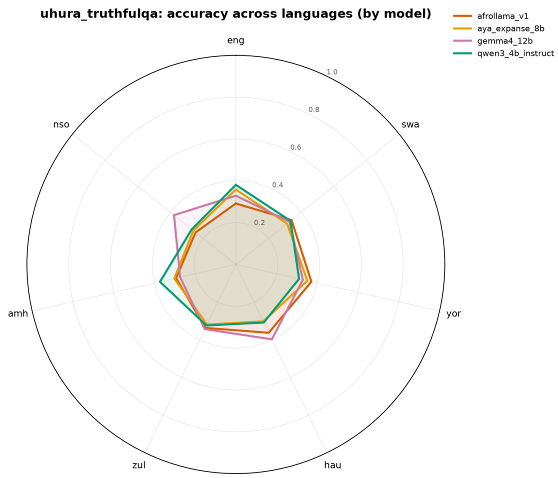


(a) Uhura-TruthfulQA

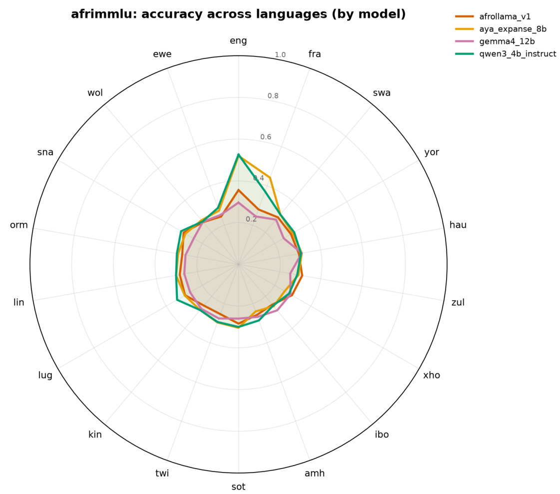


(b) AfriMMLU

Figure 2: Pooled ECE before versus after per-language temperature scaling; ECE drops to about 0.03 for all models on both datasets.

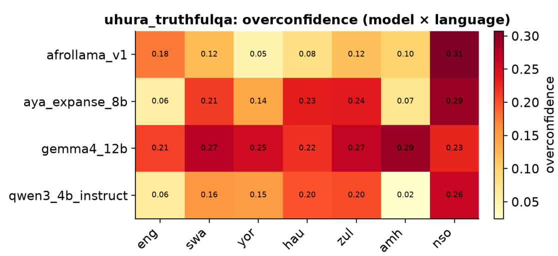


(a) Uhura-TruthfulQA accuracy by language

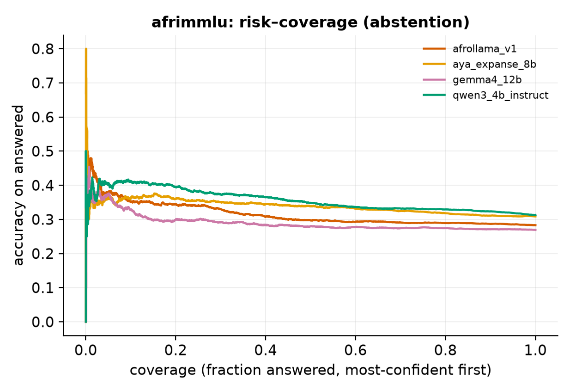


(b) AfriMMLU accuracy by language

Figure 3: Accuracy across languages by model. Long English and French spokes collapse toward chance across African languages.



(a) Uhura-TruthfulQA overconfidence by language



(b) AfriMMLU risk-coverage (abstention)

Figure 4: Overconfidence by language (left) and the accuracy-rejection trade-off under selective abstention (right); AUARC sits only modestly above baseline, reflecting a weak uncertainty signal (AUROC about 0.55).

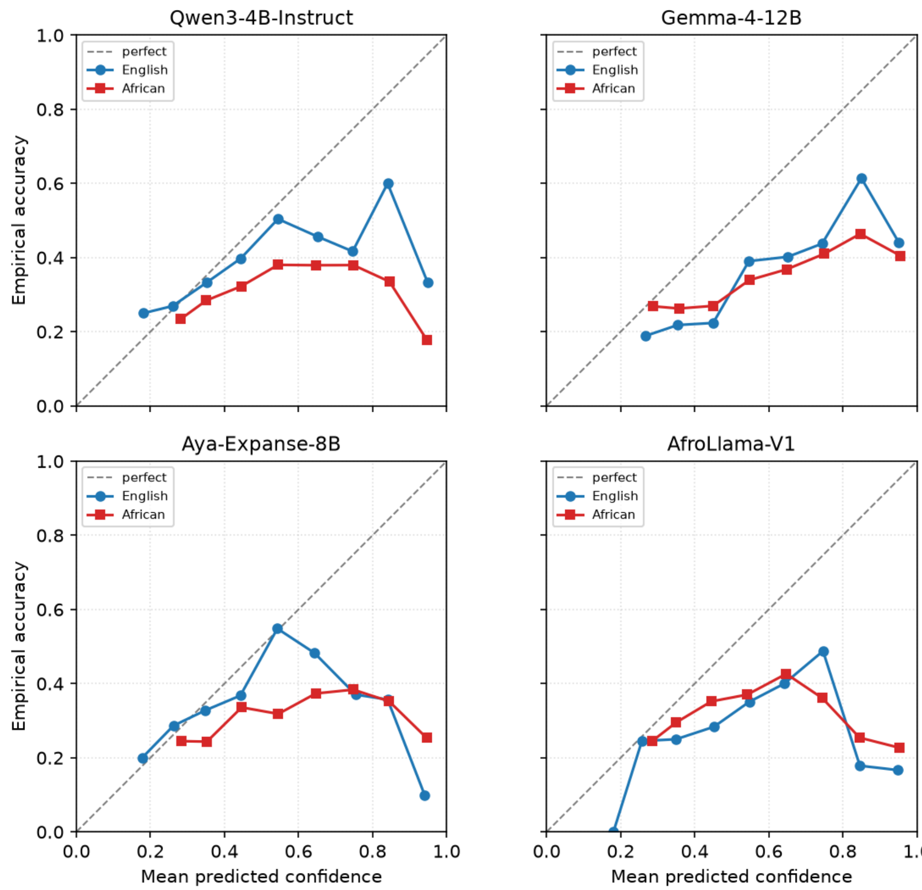


Figure 5: Reliability curves on Uhura-TruthfulQA, English versus African languages, for each model (10 confidence bins). Points on the diagonal are perfectly calibrated; points below it indicate overconfidence. The African curve sits further below the diagonal than the English curve for the general-purpose models.

Appendix G. Bootstrap confidence intervals

Table 6 reports 95% bootstrap confidence intervals (2,000 draws, stratified by language, fixed seed 42) for the four primary point estimates in Table 1 of the main paper. For each model and dataset we split items into English and African subsets, group by language, and macro-average per-language ECE and accuracy. Each bootstrap draw resamples items with replacement within each language stratum, recomputes the per-language metrics, and macro-averages; the interval is the 2.5th to 97.5th percentile of the resulting distribution. Because the point estimates here are macro-averaged across languages, they can differ by up to about 0.004 from the pooled values in Table 1 of the main paper.

Table 6: Bootstrap 95% confidence intervals for ECE and accuracy (zero-shot, 2,000 draws, stratified by language, seed 42). Each cell shows the point estimate [lower, upper].

Model	Dataset	ECE(en)	ECE(afr)	Acc(en)	Acc(afr)
Qwen3-4B	Uhura-TruthfulQA	0.072 [0.060, 0.113]	0.169 [0.160, 0.186]	0.381 [0.347, 0.413]	0.320 [0.306, 0.333]
Gemma-4-12B	Uhura-TruthfulQA	0.212 [0.182, 0.245]	0.257 [0.246, 0.273]	0.329 [0.297, 0.362]	0.342 [0.328, 0.355]
Aya-Expanse-8B	Uhura-TruthfulQA	0.075 [0.062, 0.114]	0.199 [0.187, 0.214]	0.358 [0.326, 0.392]	0.308 [0.295, 0.321]
AfroLlama-V1	Uhura-TruthfulQA	0.177 [0.148, 0.212]	0.135 [0.126, 0.151]	0.292 [0.261, 0.325]	0.325 [0.312, 0.339]
Qwen3-4B	AfriMMLU	0.062 [0.052, 0.116]	0.150 [0.147, 0.165]	0.526 [0.482, 0.572]	0.300 [0.290, 0.310]
Gemma-4-12B	AfriMMLU	0.311 [0.273, 0.354]	0.344 [0.335, 0.355]	0.296 [0.258, 0.336]	0.268 [0.259, 0.277]
Aya-Expanse-8B	AfriMMLU	0.057 [0.048, 0.113]	0.173 [0.170, 0.189]	0.520 [0.476, 0.564]	0.296 [0.286, 0.306]
AfroLlama-V1	AfriMMLU	0.225 [0.187, 0.267]	0.207 [0.201, 0.219]	0.356 [0.312, 0.398]	0.279 [0.269, 0.288]

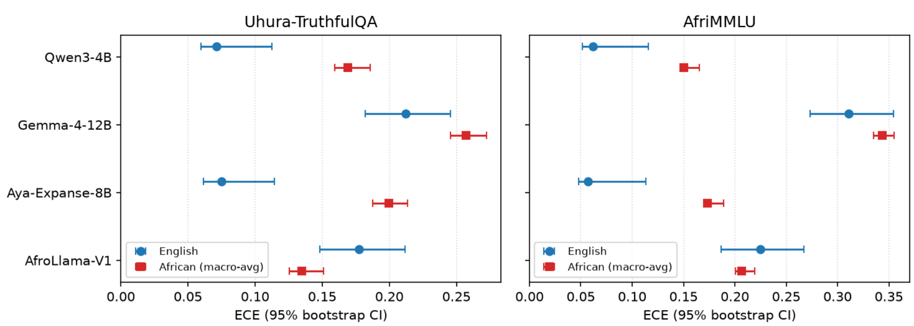


Figure 6: Bootstrap 95% confidence intervals for ECE by model, English versus African languages (markers are point estimates, bars are 95% intervals). For the general-purpose models the African interval lies entirely to the right of (above) the English interval; the intervals are disjoint.

English ECE intervals are wider because English is a single-language stratum, whereas African ECE is macro-averaged over many languages, which reduces bootstrap variance.

Appendix H. Reproducibility

All code, predictions, figures, and tables will be released in an anonymized public repository upon acceptance. The benchmarks used are publicly available: `masakhane/uhura-truthfulqa` and `masakhane/afrimmlu`.

Appendix I. LLM Usage

Claude Code was used to assist with implementing and debugging the evaluation pipeline. The research ideas, methodology, experimental design, and interpretation of results were conceptualized by the authors. All generated code, analyses, and reported results were reviewed and verified by the authors.